

Code-Layout, Readability and Reusability

Date	15 July 2026
Team ID	SWTID-2026-2749
Project Name	OptiCrop: Smart Agricultural Production Optimization engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the code base in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used Throughout the code	Yes	Uniform formatting throughout the project
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static files, models, preprocessing, and analysis
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables use descriptive names
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly indicate their purpose
5	Code Comments	Inline and block comments explain log	Yes	Important sections are documented
6	Modular Design	Code is split into reusable functions /modules	Yes	Each module performs a specific task
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code removed
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling implemented

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Prediction Module	JavaScript	Prediction Route	High
2	Random Forest Model	Scikit-learn	Crop Prediction	High
3	Flask Routing	Flask	All Web Pages	High
4	CSS Stylesheet	CSS	All HTML Pages	High
5	JavaScript	JavaScript	Navigation & UI	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized folder structure
Readability	5	Easy-to-understand code
Reusability	5	Reusable modules and functions
Documentation / Comments	5	Adequate comments and documentation
Overall Score	5/5	Well-structured Machine Learning Flask project